

Unit III

Simplification

The simplification can be asked in two ways

i) Missing numbers:

An equation can be given in the paper and we need to fill the blank in that equation given either in LHS or RHS.

for eg:

$$252 - \underline{\hspace{2cm}} + 100 = 5 \times 45 + 275$$

ii) Simplifying the equation

The second way in which the simplification questions may be asked is the direct way of giving an equation and solving it to find the answer. for eg:- $252 - 225 + 100 = ?$

Tips and tricks to solve

Always solve the approximation or simplification question following the **BODMAS Rule**

ii) Numbers which are given in a decimal format, use a rounded-off value for those numbers for eg 45.62 can be taken as 46, 22.10 can be taken as 22 etc.

* Remember the basic important formulas which can be used in such questions.

$$(a+b)^2, (a-b)^2, a^2-b^2,$$

$$a^3+b^3=(a+b)(a^2-ab+b^2)$$

$$(a+b)^3 = a^3+b^3+3ab(a+b)$$

$$(a-b)^3 = a^3-b^3-3ab(a-b)$$

Problems

$$1) \frac{(469+174)^2 - (469-174)^2}{469 \times 174}$$

Let, $469=a$ and $174=b$

$$\frac{(a+b)^2 - (a-b)^2}{ab} \Rightarrow \frac{a^2+b^2+2ab - a^2 - b^2 + 2ab}{ab} = \frac{4ab}{ab}$$

$$= 4$$

$$2) \text{ If } 2x+3y=34 \text{ \& } \frac{x+y}{y} = \frac{13}{8} \text{ find } 6x+4y$$

$$\frac{x+y}{y} = \frac{13}{8} \text{ then } y=8 \text{ and } x=5$$

$$6(5)+4(8)=30+32=62$$

$$3) \frac{(14)^2 - (1)^2}{7 \times 4}$$

$$\frac{196 - 49 \times 2}{28} = \frac{196 - 98}{28} = \frac{98}{28} = \frac{7}{2}$$

$$\Rightarrow \frac{7}{2} \text{ or } 3.5$$

$$4) \frac{8.73 \times 8.73 \times 8.73 + 4.27 \times 4.27 \times 4.27}{8.73 \times 8.73 - 8.73 \times 4.27 + 4.27 \times 4.27}$$

Let $8.73 = a$ and $4.27 = b$

$$\frac{a^3 + b^3}{a^2 - ab + b^2} = \frac{(a+b)(a^2 - \cancel{ab} + b^2)}{a^2 - \cancel{ab} + b^2} = a+b = 8.73 + 4.27 = 13$$

$$5) 7\frac{1}{2} - \left[2\frac{1}{4} \div \left\{ 1\frac{1}{4} - \frac{1}{2} \left(\frac{1}{2} - \frac{1}{3} - \frac{1}{6} \right) \right\} \right]$$

$$\frac{15}{2} - \left[\frac{9}{4} \div \left\{ \frac{5}{4} - \frac{1}{2} \left(\frac{3}{2} - \frac{1}{3} - \frac{1}{6} \right) \right\} \right]$$

first solve $\left(\frac{3}{2} - \frac{1}{3} - \frac{1}{6} \right) \Rightarrow \frac{6-2-1}{6} = \frac{3}{6} \Rightarrow 1$

$$\frac{15}{2} - \left[\frac{9}{4} \div \left\{ \frac{5}{4} - \frac{1}{2} (1) \right\} \right]$$

$$\frac{15}{2} - \left[\frac{9}{4} \div \left\{ \frac{3}{4} \right\} \right] \Rightarrow \frac{15}{2} - \left[\frac{9^3}{4} \times \frac{4}{3} \right]$$

$$\frac{15}{2} - 3 \Rightarrow \frac{9}{2} \text{ or } 4.5$$

6) A man has ~~Rs. 480~~ Rs. 480 in the denomination of one rupee notes, five rupee notes and ten rupee notes. the no. of notes of each denomination is equal. what is the total no. of notes that he has?

$$1x + 5x + 10x = 480$$

$$16x = 480$$

$$x = 30$$

Rs. 1 note = 30

Rs. 5 note = 30

Rs. 10 note = 30

$$\boxed{\text{Total} = 90}$$

7) There are two examination rooms A and B.

If 10 students are sent from A to B, then the no. of students in each room is the same. If

20 candidates are sent from B to A, then the no. of students in A is double the no. of students in B. find the no. of students in room A

Room A

Room B

$$x - 10 = y + 10 \quad \dots \quad (i)$$

$$x + 20 = 2(y - 20) \quad \dots \quad (ii)$$

$$(i) \quad x - y = 20 \rightarrow \textcircled{1}$$

$$(ii) \quad x - 2y = -60 \rightarrow \textcircled{2}$$

$$\textcircled{1} \times 2 \Rightarrow 2x - 2y = 40$$

$$\textcircled{2} \quad \underline{x - 2y = -60}$$

$$x = 100 //$$

8) The price of 10 chairs is equal to that of 4 tables. The price of 15 chairs and 2 tables together is Rs. 4000. find the total price of 12 chairs and 3 tables

Let, chairs = x & tables = y

$$10x = 4y \rightarrow \textcircled{1}$$

$$15x + 2y = 4000 \rightarrow \textcircled{2}$$

$$12x + 3y = ?$$

If $10x = 4y$, then $5x = 2y$

$$\textcircled{2} \rightarrow 15x + 5x = 4000$$

$$x = 200$$

$$y = 500$$

$$\begin{array}{l} 12x + 3y \\ 12(200) + 3(500) \\ 3900 // \end{array}$$

9) A fires ~~8~~ 5 shots to B's 3 but A kills only once in 3 shots while B kills only in 2 shots. When B has missed 27 times, A has killed.

Let the total no. of shots be x . Then,

Shots fired by A = $\frac{5}{8}x$ and B = $\frac{3}{8}x$

Killing shots by A = $\frac{1}{3}$ of $\frac{5}{8}x = \frac{5}{24}x$

Shots missed by B = $\frac{1}{2}$ of $\frac{3}{8}x = \frac{3}{16}x$

$$\frac{3x}{16} = 27 \text{ or } x = \frac{27 \times 16}{3} = 144.$$

$$\text{Birds killed by A } \frac{5x}{24} = \frac{5}{24} \times 144 = 30$$

10) To fill a tank, 25 buckets of water is required. How many buckets of water is required to fill the same tank if the capacity of the bucket is reduced to two-fifth of its present

Capacity of 1 bucket = x

Capacity of tank = $25x$

New capacity of bucket = $\frac{2}{5}x$

$$\text{Req. no. of buckets} = \frac{25x}{\frac{2x}{5}} \Rightarrow \left[25x \times \frac{5}{2x} \right]$$

$$= 62.5$$

Surds and Indices

Laws of Indices

$$a^m \times a^n = a^{m+n}$$

$$a^m / a^n = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n$$

$$\left(\frac{a}{b}\right)^n = a^n / b^n$$

$$a^0 = 1$$

Laws of Surds

$$\sqrt[n]{a} = a^{1/n}$$

$$\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$$

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$(\sqrt[n]{a})^n = a$$

$$\sqrt[n]{n\sqrt[n]{a}} = \sqrt[n]{a}$$

$$(\sqrt[n]{a})^m = \sqrt[n]{a^m}$$

Problems

1) $(17)^{3.5} \times (17)^? = 17^8$

$$3.5 + x = 8 \Rightarrow x = 4.5$$

2) If $\left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{x-3}$ then the value of x

$$\left(\frac{a}{b}\right)^{x-1} = \left(\frac{a}{b}\right)^{3-x} \Rightarrow x-1 = 3-x \Rightarrow \boxed{x=2}$$

3) Given that $10^{0.48} = x$, $10^{0.70} = y$ & $x^z = y^2$
then the value of z is close to

$$x^z = y^2$$
$$10^{0.48z} = 10^{(0.70)2}$$

$$0.48z = 1.4$$

$$z = \frac{1.4}{0.48} = \frac{140}{48}$$

$$z = \frac{35}{12}$$

4) $5^a = 3125$ then the value of $5^{(a-3)}$

$$5^a = 5^5 \Rightarrow a = 5$$

$$5^{(5-3)} = 5^2 = 25$$

5) If $3^{(x-y)} = 27$ and $3^{(x+y)} = 243$ then $y = ?$

$$3^{(x-y)} = 3^3 \text{ and } 3^{(x+y)} = 3^5$$

$$x - y = 3$$

$$x + y = 5$$

$$\underline{\underline{x = 4}}$$

$$y = 1$$

6) $(256)^{0.16} \times (256)^{0.09} = ?$

$$256^{0.16+0.09} \Rightarrow 256^{0.25}$$

$$= (4^4)^{25/100} \Rightarrow (4^4)^{1/4} = 4$$

7) The value of $[(10)^{150} \div (10)^{146}]$

$$10^{150-146} = 10^4 \Rightarrow 10,000$$

8) $(25)^{7.5} \times 5^{2.5} \div (125)^{1.5} = ?$

$$(5^2)^{7.5} \times 5^{2.5} \div (5^3)^{1.5} = 5^x$$

$$\frac{5^{15} \times 5^{2.5}}{5^{4.5}} = 5^x \Rightarrow 5^{15+2.5-4.5} = 5^x \Rightarrow 5^{13} \Rightarrow \boxed{x=13}$$

9) $(0.04)^{-1.5} = ?$

$$\left(\frac{4}{100}\right)^{-1.5} \Rightarrow \left(\frac{100}{4}\right)^{1.5} \Rightarrow (25)^{1.5} = (5^2)^{1.5} = 5^3 \Rightarrow 125$$